

● EASY

● ALGEBRA

Linear inequalities in one or two variables

01

10 questions · ~15 min

Q1

ALGEBRA

Leo goes to a packing store to buy containers and tape. Leo has \$ 15. Each container costs \$ 1.87 and each roll of tape costs \$ 2.40. Which inequality represents the relationship between the number of containers, c , and the number of rolls of tape, t , Leo can buy?

- A. $1.87c + 2.40t \leq 15$
- B. $1.87c + 2.40t \geq 15$
- C. $2.40c + 1.87t \leq 15$
- D. $2.40c + 1.87t \geq 15$

$$2l + 2w \leq 27$$

A rectangle has length l and width w . The inequality gives the possible values of l and w for which the perimeter of this rectangle is less than or equal to 27. Which statement is the best interpretation of $l, w = 8, 3$ in this context?

- A. If the rectangle has length 3 and width 8, its perimeter is less than or equal to 27.
- B. If the rectangle has length 8 and width 3, its perimeter is less than or equal to 27.
- C. If the rectangle has length 3 and width 8, its perimeter is greater than or equal to 27.
- D. If the rectangle has length 8 and width 3, its perimeter is greater than or equal to 27.

For a 3-week period in a town in Illinois, the lowest recorded temperature was 31 degrees Fahrenheit $^{\circ}\text{F}$ and the highest recorded temperature was 67°F . Which inequality is true for any recorded temperature t , in $^{\circ}\text{F}$, in this town for this 3-week period?

- A. $t \geq 98$
- B. $t \geq 67$
- C. $31 \leq t \leq 67$
- D. $t \leq 31$

Q4

ALGEBRA

Valentina bought two containers of beads. In the first container 30% of the beads are red, and in the second container 70% of the beads are red. Together, the containers have at least 400 red beads. Which inequality shows this relationship, where x is the total number of beads in the first container and y is the total number of beads in the second container?

A. $0.3x + 0.7y \geq 400$

B. $0.7x + 0.3y \leq 400$

C. $\frac{x}{3} + \frac{y}{7} \leq 400$

D. $30x + 70y \geq 400$

Q5

ALGEBRA

For a literature class, a student's essay needs to have no more than 5,000 words. If a student's essay already has 3,026 words, what is the maximum number of additional words the student can include in the essay?

A. 1,974

B. 3,026

C. 5,000

D. 8,026

Q6

ALGEBRA

Julissa needs at least 100 hours of flight time to get her private pilot certification. If Julissa already has 86 hours of flight time, what is the minimum number of additional hours of flight time Julissa needs to get her private pilot certification?

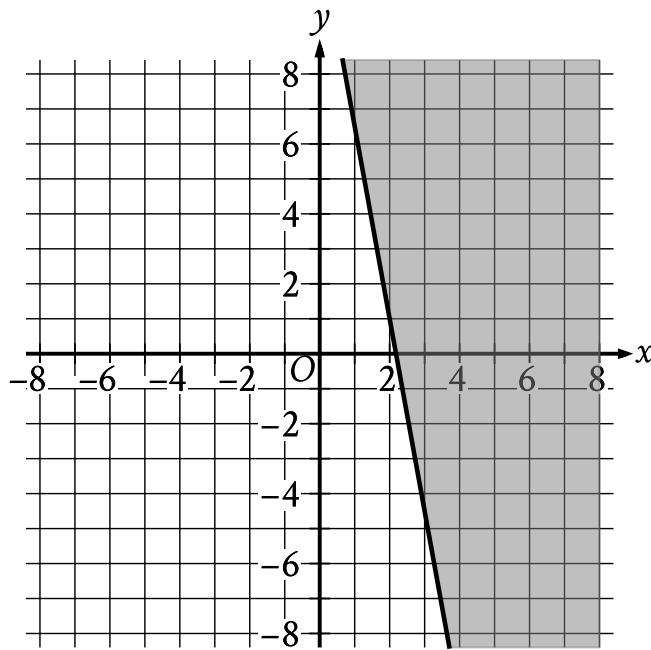
- A. 14
- B. 76
- C. 86
- D. 186

Q7

ALGEBRA

A bakery sells trays of cookies. Each tray contains at least 50 cookies but no more than 60. Which of the following could be the total number of cookies on 4 trays of cookies?

- A. 165
- B. 205
- C. 245
- D. 285



- The boundary of the inequality is a solid line.
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (1.0 comma 6.5)
 - (3.0 comma negative 4.5)
- The area above and to the right of the boundary is shaded.

The shaded region shown represents solutions to an inequality. Which ordered pair x, y is a solution to this inequality?

- A. 0, -4
- B. 0, 4
- C. -4, 0

D. 4,0

Q9

ALGEBRA

Ty set a goal to walk at least 24 kilometers every day to prepare for a multiday hike. On a certain day, Ty plans to walk at an average speed of 4 kilometers per hour. What is the minimum number of hours Ty must walk on that day to fulfill the daily goal?

A. 4

B. 6

C. 20

D. 24

For a party, 50 dinner rolls are needed. Dinner rolls are sold in packages of 12. What is the minimum number of packages that should be bought for the party?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.